In[1]:=
```
subsβ = {β → 2 Mw + 2 Jw / (Rw^2) + Mb};
subsα = {α → Jb * β + (2 Mb * L^2) (Mw + Jw / (Rw^2))};
```

In[3]:=
```
A =
  {{0, 1, 0, 0}, {0, 2 Km * Ke (Mb * L * Rw - Jb - Mb * (L^2)) / (α * R * (Rw^2)), Mb * g * L^2 / α, 0}, {0, 0, 0, 1},
    {0, 2 Km * Ke (Rw * β - Mb * L) / (α * R * Rw^2), Mb * g * L * β / α, 0}} /. subsα /. subsβ;
```

In[4]:=
```
MatrixForm [A]
```

Out[4]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{2\,Ke\,Km\left(-Jb-L^2\,Mb+L\,Mb\,Rw\right)}{R\left(2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)\right)Rw^2} & \frac{g\,L^2\,Mb}{2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2\,Ke\,Km\left(-L\,Mb+\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)Rw\right)}{R\left(2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)\right)Rw^2} & \frac{g\,L\,Mb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)}{2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)} & 0 \end{pmatrix}$$

In[5]:=
```
B = {{0}, {2 Km (Jb + Mb * (L^2) - Mb * L * Rw) / (Rw * R * α)}, {0}, {2 Km (Mb * L - Rw * β) / (Rw * R * α)}} /. subsα /. subsβ;
```

In[6]:=
```
model = StateSpaceModel [{A, B, {{0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}, {{0}, {0}, {0}}},
  SystemsModelLabels → {{"Voltage"}, {"vel", "angle", "ang_vel"}, {"pos", "vel", "angle", "ang_vel"}}]
```

Out[6]=

					Voltage
pos	0	1	0	0	0
vel	0	$\frac{2\,Ke\,Km\left(-Jb-L^2\,Mb+L\,Mb\,Rw\right)}{R\left(2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)\right)Rw^2}$	$\frac{g\,L^2\,Mb}{2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)}$	0	$\frac{2\,Km\left(Jb+L^2\,Mb-L\,Mb\,Rw\right)}{R\left(2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)\right)Rw}$
angle	0	0	0	1	0
ang_vel	0	$\frac{2\,Ke\,Km\left(-L\,Mb+\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)Rw\right)}{R\left(2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)\right)Rw^2}$	$\frac{g\,L\,Mb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)}{2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)}$	0	$\frac{2\,Km\left(L\,Mb-\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)Rw\right)}{R\left(2\,L^2\,Mb\left(Mw+\frac{Jw}{Rw^2}\right)+Jb\left(Mb+2\,Mw+\frac{2\,Jw}{Rw^2}\right)\right)Rw}$
vel	0	1	0	0	0
angle	0	0	1	0	0
ang_vel	0	0	0	1	0

```
Data = {Rw → 0.2, Mw → 4, Jw → 0.07, L → 1.8, Mb → 120, Jb → 60, g → 9.8, Km → 0.069, Ke -> 0.083, R → 0.5 / 6};
```

In[8]:=
```
minModel = SystemsModelDelete [model, None, None, 1] /. Data
```

Out[8]=

				Voltage
vel	-0.11275	0.308242	0	0.271686
angle	0	0	1	0
ang_vel	-0.0527333	22.5188	0	0.127068
vel	1	0	0	0
angle	0	1	0	0
ang_vel	0	0	1	0

In[9]:=
```
gains = StateFeedbackGains [minModel, {-10, -5, -1.5}]
```

Out[9]= {{-12.7528, 747.778, 156.231}}

In[10]:=
```
rlen = L /. Data;
```